

STA 4350 FINAL Project

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1 Introduction

Within a capitalistic-based society, it is important for many individuals to create new products, services, or opportunities that benefit the surrounding community. Banks often give customers opportunities to borrow money so that the customer has the funds for their respective desires. However, Banks are businesses that still need to continue existing by making profits; they need to analyze the risk of giving a customer a loan. Our goal is to analyze data from 5,000 customers who received a loan from a bank in order to identify the risk of customers defaulting on their loan. We will do some exploratory data analysis, create different logistic regression models using statistical inference, and determine the best of those candidate models that best model customer default risk of a financial loan.

2 Data

Looking at our banking customers dataset, we have a total of 255,347 banking customers who received some sort of loan, and we have 18 different variables that describe the customers at the time when they received the loan. However, we chose to reduce the observation amount to 5000 individuals to ensure the use of statistical inference within the project. We have the customer's identification number within the banking system. We have the age of the customer in years. The amount of income in U.S dollars they made a year. We have their FICO credit score of the customer. How many months the customer had already had at the time. How many lines of credit did the customer have access to. The interest rate of the loan they received from the bank. As well as the loan term in months. The Debt-to-Income ratio of the loan recipient. The highest level of education, which is classified by high school level, bachelor's, master's, and PHD. Employment type has 4 levels: self-employment, unemployment, part-time, or full-time. Marital Status of being single, married, or divorced. If they have a mortgage at the time, and if they have dependents. The reason of the loan being auto, business, education, home or other. If the borrower had a cosigner at the time and if they defaulted on loan. For my data cleaning steps, I simply ensured that we did not have missing entries, which we did not. Also,

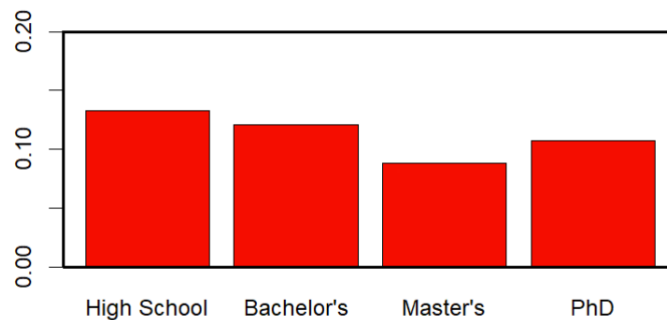
looking for any observations that were outliers or flagged for missing entries in the process of data entry but there was none that I saw.

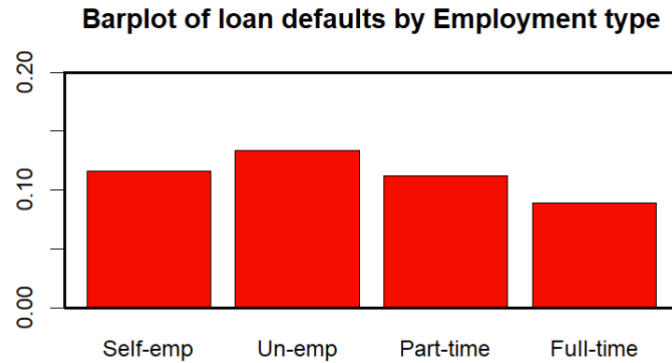
Variable	Min	Q1	Median	Q3	Max
Age (years)	18	31	44	57	69
Income (U.S dollars)	150023	49057	83298	116537	149963
Loan Amount (U.S dollars)	5081	64193	126716	188307	249996
Credit Score	300	438	582	712	849
Months Employed	0	29	60	89	119
Number of Credit Lines	1	1	2	3	4
Interest Rate	2.030	7.808	13.540	19.410	24.990
Loan Term	12	24	36	48	60
Debt-to-Income Ratio	0.10	.30	0.50	0.70	0.90

Table 1: Five-number summary for numeric variables

Education	Count	Employment	Count	Marital Status	Count
High School	763	Full-Time	784	Divorced	969
Bachelors	744	Part-Time	716	Married	1013
Master's	718	Self-Employed	757	Single	1018
PHD	775	Unemployed	743		
				Loan Purpose	Count
HasMortgage	Count	HasDependents	Count	Auto	582
No	1519	No	1471	Business	592
Yes	1481	Yes	1529	Education	579
				Home	595
				Other	652
HasCosigner	Count				
No	1551	DEFAULT			
Yes	1449	No	2663	Yes	337

Barplot of loan defaults by Education levels

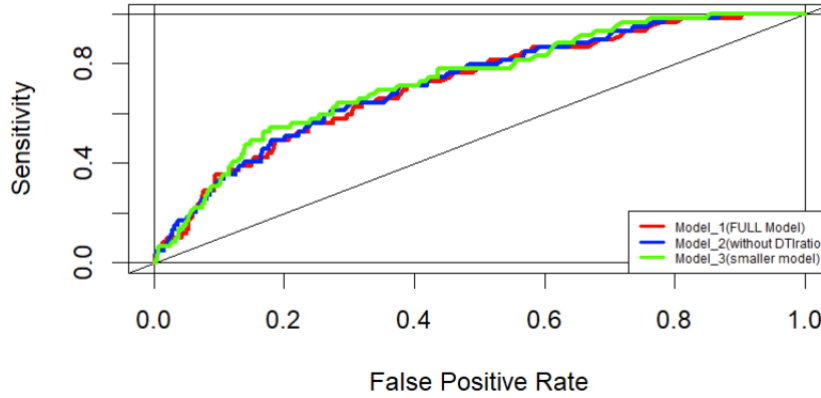




3 Model Selection

The process of selecting my potential candidate model went as followed. First, I ran the logistic regression model with all the 17 featured variables. I examined the Wald z-test for every variable to see which variable were not associate with the risk of defaulting on their loan given every other variable was fixed. These variables were as listed LoanTerm, DTI-ratio, had a Mortgage,has mortgage status, marital status Loan Purpose, had cosigner, and Employment type variables. However, the variable DTI-ratio had p-values that were pretty close to 0.05 so I decided to make our second candidate model all the variables except that variable. Our third candidate model is just the full model removing every featured variable that was not associated with the risk of default using the Wald z-test, except for the variable employment type. Now have 3 candidate model from our statistical inference, I decided to test their prediction accuracy score using K=10 fold cross validation. So what I did was randomly sampled 20 percent of our 3000 observation to use as our testing dataset. The other 80 percent I trained the 3 candidate logistic regression models on. With K=10 fold cross validation we are taking 80 percent of 3000 samples we have dividing it into 10 folds where we are going to train the model on 9 of the folds at a time. After training the model on 9 different combination of folds where each fold that not used for training was used for the validation set. We then looked at the summary for all 3 models to get that the full model had an AIC of 1369, the model without just DTI-ratio had an AIC of 1370, and 1368.4. Then, performing the Likelihood ratio test with the subset model 2 compared to the full model we get a p-value of 0.08083 which means that we have insufficient evidence to fail to reject the null hypothesis. The model 2 (subset model) is just as good as our full model. Doing the Likelihood ratio test with model 3 compared to model 2 we get a p-value of 0.0591. We have insufficient evidence to fail to reject the null hypothesis, thus model 3 is just as good as a model compared to model 2. Thus, so far our model 3 is the best model in terms of statistical inference

according to our AIC value and likelihood ratio test. Now to determine what model is best in terms of prediction we will use the average prediction rate, ROC curves and the area under the curve. Assuming we use 0.5 as our threshold cutoff, the testing dataset when comparing the average prediction of a customer defaulting on their loan model 1 got 90.33 percent accuracy, model 2 got 0.9 percent accuracy and model 3 got 89.83 percent accuracy.



Then after plotting the ROC curve for each of our candidate models it seems that model 3 with the green line seems to have a bigger ROC than model 1 and 2. However, I can not differentiate the next best model according to the ROC plot. So using a package called pROC we get that the area under the curve for model 1 was 0.7157, model 2 was 0.7222, and 0.7309. Model 3 has the highest AUC which is the probability that it will rank probability values for a random customer that defaulted on their loan higher than the probability for a customer who did not default on their loan. Thus based on statistic inference and prediction metrics, model 3 is the best out of our 3 candidate models.

4 Results

Using model 3 as our best candidate model, we can see the probability of customers in different scenarios. For example, say if we gave a loan to a 25-year-old, with an income of 85000 dollars, getting a loan amount of 200000 dollars, with 750 credit score, 3 credit lines, 24 months employed, 10 percent interest rate on the loan, with a master's degree, full time job, and does not have any dependents. The probability in this specific case of the banking customer who received a loan of them defaulting is 13.24 percent. We could consider these individuals to have a low risk of defaulting on their loan. We can interpret the odds for our variables within our logistic regression model and give their 95 percent confidence levels. We are 95 percent sure that when the age of the banking customer who received a loan increases by 10 years the odds of them defaulting on their loan is lower between 41.85 percent and 28.84

percent, assuming all other variables are fixed. We are 95 percent sure that when the Income of the banking customer who received a loan increases by 10,000 dollars, the odds of them defaulting on their loan is lower between 11.89 percent and 5.21 percent, assuming all other variables are fixed. We are 95 percent sure that when the Loan amount of the banking customer who received a loan increases by 50,000 dollars the odds of them defaulting on their loan increase between 7.92 percent and 31.76 percent, assuming all other variables are fixed. We are 95 percent sure that when the Credit-Score of the banking customer who received a loan increases 50,000 dollars, the odds of them defaulting on their loan decrease between 21.28 percent and 5.87 percent, assuming all other variables are fixed. We are 95 percent sure that when the months employed of the banking customer who received a loan increased 2 years, the odds of them defaulting on their loan decreased between 26.5 percent and 10.45 percent, assuming all other variables are fixed. We are 95 percent sure that when the months employed of the banking customer who received a loan increased by 2 years, the odds of them defaulting on their loan decreased by 1.1 percent and increased by 27.59 percent, assuming all other variables are fixed. We are 95 percent sure that when the interest rate of the banking customer who received a loan increased by 4 percent the odds of them defaulting on their loan increased between 24.46 percent and 48.30 percent, assuming all other variables are fixed. We are 95 percent sure that when the highest education level obtained of the banking customer who received a loan is a Master's degree, compared to a High school diploma ,the odds of them defaulting on their loan decreased between 62.17 percent and 13.57 percent, assuming all other variables are fixed. We are 95 percent sure that when the job type of the banking customer that received a loan is a full-time job compared to a self-employed job, the odds of them defaulting on their loan decreased between 53.28 percent and increased 5.56 percent, assuming all other variables are fixed. We are 95 percent sure that the banking customer who has dependents compared to not having dependents, the odds of them defaulting on their loan decreased between 50.16 percent and 12.11 percent, assuming all other variables are fixed.

5 Summary

Our key finding was that the best logistic regression model for modeling the risk of customers defaulting on their loans using our banking dataset, with originally 18 different featured variables within the data, was a subset of the original data. This logistic model was comprised of using the variables age, income, loan amount, credit score, months of employment, interest rate of the loan, highest education level, job type, and if the banking customers have dependents. We selected this model using statistical inference and prediction metrics using Wald z-test, training/testing datasets using K-fold cross-validation, AIC values, Likelihood ratio test, ROC curves, and Area under the curves.

{Appendix,Rcode}

```
- ds <- read_csv(archive_read("archieive_bank.zip", file = "Loan_default.csv"))
- set.seed(4350)
- subset_data <- sample(nrow(ds),3000)
- ds<- ds[subset_data,]
- logistic_1 <- glm(Default~., data=ds,family=binomial)
- summary(logistic_1)
- logistic_2 <- glm(Default~.-DTIRatio, data=ds,family=binomial) # ratio
- summary(logistic_2)
- # AIC: 1877.3
- anova(logistic_2,logistic_1, test="LRT")
- logistic_3 <-
  glm(Default~.-LoanTerm-LoanPurpose-HasCoSigner-MaritalStatus-DTIRatio-ds$HasMortgage,
  data=ds, family=binomial)
- summary(logistic_3)
- anova(logistic_3,logistic_2,test="LRT") # 0.1511
```

```
n_total <- nrow(ds)
n_test <- 0.2*n_total
n_ds<- n_total-n_test
k<- 10
ds_fold <- n_ds/k
set.seed(4350)
test_i <- sample(n_total, n_test)
ds_test<- ds[test_i,]
ds_cv <- ds[-test_i,]
perm <- sample(n_ds,n_ds)
pi_hat_1= pi_hat_2 = pi_hat_3 = numeric(n_ds)
```

```
for(j in 1:k){
```

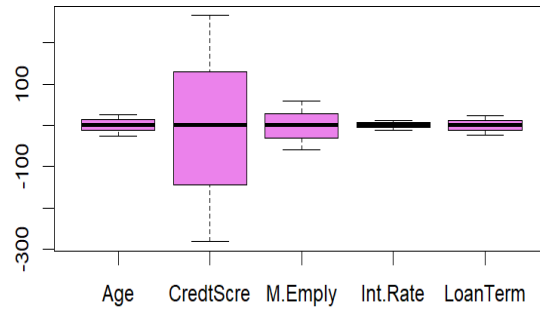
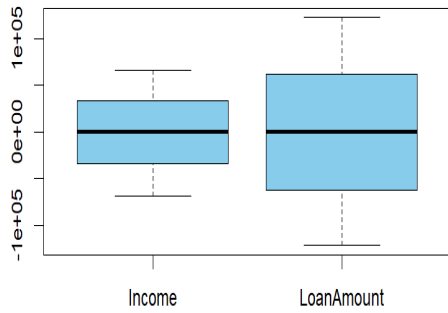
```
  fold <- perm[1:ds_fold + (j-1)*ds_fold]
```

```
  mlr_1<- glm(Default~., data=ds_cv,family=binomial,subset=-fold)
  mlr_2 <- glm(Default~.-DTIRatio, data=ds_cv,family=binomial,subset=-fold)
  mlr_3 <- glm(Default~.-LoanTerm-LoanPurpose-HasCoSigner-MaritalStatus-DTIRatio-HasMortgage,
  data=ds_cv, family=binomial,subset=-fold)
  pi_hat_1[fold]<- predict(mlr_1, newdata=ds_cv[fold,], type="response")
  pi_hat_2[fold]<- predict(mlr_2, newdata=ds_cv[fold,], type="response")
  pi_hat_3[fold]<- predict(mlr_3, newdata=ds_cv[fold,], type="response")
```

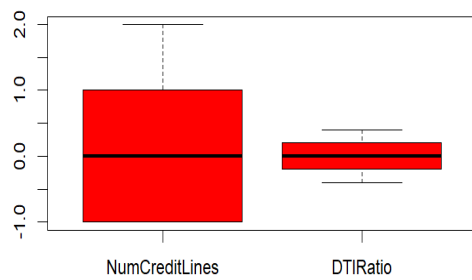
```
- scenario_1 <- coef(mlr_3)[("(Intercept)"] + coef(mlr_3)[("Age"]*25 + coef(mlr_3)[("Income"]*85000 +
  coef(mlr_3)[("LoanAmount"]*200000 + coef(mlr_3)[("CreditScore"]*750 +
  coef(mlr_3)[("NumCreditLines"]*3 + coef(mlr_3)[("MonthsEmployed"]*24 +
  coef(mlr_3)[("InterestRate"]*10 + coef(mlr_3)[("EducationBachelor's"]*0 +
  coef(mlr_3)[("EducationMaster's"]*1 + coef(mlr_3)[("EducationPhD"]*0 +
  coef(mlr_3)[("EmploymentTypeUnemployed"]*0 + coef(mlr_3)[("EmploymentTypePart-time"]*0 +
  coef(mlr_3)[("EmploymentTypeFull-time"]*1 + coef(mlr_3)[("HasDependentsYes"]*0
```

```
- prob_scen1<- (1/(1+ exp(-(scenario_1))));prob_scen1
```

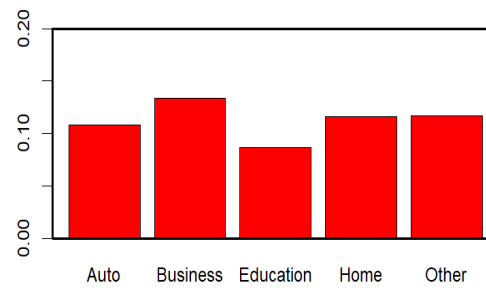
Aligned Boxplots of Featured variables



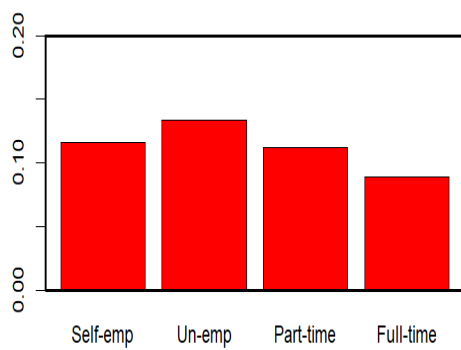
Aligned Boxplots of Featured variables



Barplot of loan defaults by Loan Purpose



Barplot of loan defaults by Employment type



Barplot of loan defaults by Education levels

